

Qualification Pack



Dairy Processing and Development Supervisor

QP Code: CUTM/AGR/Q1201

Version: 1.0

NSQF Level: 5

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Qualification Pack

Contents

CUTM/AGR/Q1201: Dairy Processing and Development Supervisor	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
CUTM/AGR/N1201: Understand and Execute Collection, Validation, Processing and Storage of Raw Milk	5
CUTM/AGR/N1202: Apply Dairy Starters and Execute Microbiological Analysis of Milk and Milk Products	11
CUTM/AGR/N1203: Process Different Types of Dairy Products	16
CUTM/AGR/N1204: Perform Chemical Quality Testing of Raw Milk, Processed Milk and Milk Products	22
CUTM/VSQ/N7804: Ensure organizational health, safety and security for self and others	28
DGT/VSQ/N0102: Employability Skills (60 Hours)	33
Assessment Guidelines and Weightage	40
<i>Assessment Guidelines</i>	40
<i>Assessment Weightage</i>	42
Acronyms	43
Glossary	45

Qualification Pack

CUTM/AGR/Q1201: Dairy Processing and Development Supervisor

Brief Job Description

The Individual is able to understand the technologies of raw milk processing operations, get hands on experience to prepare different dairy products, microbiological analysis and role of dairy starters in manufacturing fermented dairy products, monitor safety and analyze chemical quality of milk and milk products.

Personal Attributes

A Dairy Processing and Development Supervisor should be quality-focused, responsible, and attentive to detail, with strong technical knowledge of dairy operations. They must demonstrate leadership in managing teams, problem-solving ability in handling processing challenges, and adaptability to modern dairy technologies. Good communication, planning, hygiene consciousness, and commitment to food safety and efficiency are essential.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N1201: Understand and Execute Collection, Validation, Processing and Storage of Raw Milk](#)
2. [CUTM/AGR/N1202: Apply Dairy Starters and Execute Microbiological Analysis of Milk and Milk Products](#)
3. [CUTM/AGR/N1203: Process Different Types of Dairy Products](#)
4. [CUTM/AGR/N1204: Perform Chemical Quality Testing of Raw Milk, Processed Milk and Milk Products](#)
5. [CUTM/VSQ/N7804: Ensure organizational health, safety and security for self and others](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Dairy Products and Processing
Occupation	Dairy Farm Management
Country	India



Qualification Pack

NSQF Level	5
Credits	19
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (in Biology/ Microbiology) with NA of experience OR Completed 1st year of UG (UG Certificate) (in Biology/ Microbiology) with 1.5 years of experience relevant Experience OR 12th grade Pass with 3 Years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant Experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-05-AG-04289-2025-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/AGR/N1201: Understand and Execute Collection, Validation, Processing and Storage of Raw Milk

Description

The NOS "Understand and execute collection, validation, processing and storage of raw milk" under the qualification Dairy Processing and Development Supervisor (NSQF Level 5) focuses on equipping learners with the knowledge and skills to manage hygienic milk collection and processing operations effectively. It involves implementing clean milk production practices, understanding milk pricing systems, ensuring quality validation through physical, chemical, and microbiological parameters, and carrying out standard operating protocols for separation, pasteurization, homogenization, chilling, and storage of milk. Additionally, the NOS emphasizes thorough documentation and adherence to safety and quality protocols across all processes.

Scope

The scope covers the following :

- This NOS covers the end-to-end operations of raw milk handling, starting from hygienic milk production at the farm level to systematic procurement, quality validation, processing, and appropriate storage. It includes implementation of milk pricing systems, platform quality tests, usage of dairy equipment (e.g., separators, pasteurizers, chillers), and adherence to SOPs to maintain milk quality. It emphasizes hygienic practices, accurate documentation, and safety protocols to ensure high-quality raw milk is delivered for further processing into dairy products.

Elements and Performance Criteria

Hygienic milk production and milk pricing policies

To be competent, the user/individual on the job must be able to:

- PC1.** Milk collection and pricing system based on fat content, volume, total milk solids, species of milk animals and cost of milk production
- PC2.** Clean milk production practices in dairy farm
- PC3.** Hygienic production of milk by inspecting raw milk for fulfilment to physical, chemical, and microbiological parameters.
- PC4.** Milk preservation system.

Milk procurement and processing operations with complete documentation

To be competent, the user/individual on the job must be able to:

- PC5.** Sampling and lactometer reading of procured milk to determine quality following standard operating protocols.
- PC6.** Platform tests (COB, alcohol, sediment, organoleptic evaluation etc.) related to appearance, smell, acidity, consistency, color, temperature etc.
- PC7.** Filtration and checking microbial quality of milk according to standard protocol.
- PC8.** Process of separation through separator to separate cream from milk according to standard protocol.

Qualification Pack

- PC9.** Homogenization of milk in homogenizer to obtain desirable size of milk fat globule according to standard protocol.
- PC10.** Milk pasteurization according to standard protocol.
- PC11.** Milk cooling in chilling tank until packing.
- PC12.** Standardized and flavored milks (fruit, chocolate etc.) following standard protocol.
- PC13.** CIP cleaning of valves, pipes and pipeline circuits with proper process operations after product manufacture.
- PC14.** Documentation of processed milk related to parameters analyzed, milk storage conditions, process of analysis, equipment usage etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** pricing systems based on factors like fat content, milk solids, animal species, volume, and cost of milk production essential for rural procurement operations and fair trade.
- KU2.** pricing systems based on factors like fat content, milk solids, animal species, volume, and cost of milk production essential for rural procurement operations and fair trade.
- KU3.** how to assess physical, chemical, and microbiological parameters of raw milk, ensuring compliance with regulatory and quality standards before processing
- KU4.** preservation systems like chilling and refrigeration to prevent microbial spoilage and maintain raw milk quality during storage and transport.
- KU5.** the familiarity with lactometer readings, sampling procedures, and platform tests (COB, alcohol, sediment, organoleptic evaluations) for initial validation of milk quality at collection points.
- KU6.** the unit operations like clarification, separation (cream removal), homogenization (reducing fat globule size), and pasteurization (thermal processing) according to SOPs..
- KU7.** the preparation of standardized and flavored milk (e.g., fruit, chocolate) and the variations in milk types as per consumer or industrial demand.
- KU8.** CIP methods for cleaning valves, pipes, and circuits using alkali/acid and hot water post-production critical for hygiene and operational safety.
- KU9.** concepts and SOPs for using key dairy equipment such as milk analyzers, homogenizers, pasteurizers, chillers, and cream separators, including calibration and handling procedures.
- KU10.** proficiency in maintaining detailed records of milk quality parameters, processing conditions, storage data, and equipment usage to ensure traceability and support food safety audits.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply clean milk production practices at the dairy farm level.
- GS2.** Interpret milk pricing policies based on fat content, milk solids, animal species, and seasonal variations and Coordinate with rural milk collection centers and producers for efficient procurement.

Qualification Pack

- GS3.** Conduct physical tests such as lactometer reading and platform tests (COB, alcohol, organoleptic, sediment tests). Perform microbial quality checks and standard protocols for cream separation, homogenization, and pasteurization.
- GS4.** Familiarity with calibrated dairy equipment (weighing balance, milk analyzer, cream separator, homogenizer, pasteurizer, chiller). Follow standard operating procedures (SOPs) for all processing steps to ensure consistent quality.
- GS5.** Maintain detailed documentation for milk processing parameters, storage conditions, and test results. Ensure traceability and transparency in milk collection and handling processes.
- GS6.** Apply methods to prevent spoilage during transportation and short-term storage.
- GS7.** Identify potential quality issues (e.g., adulteration, microbial contamination) and take corrective action. Troubleshoot process deviations during milk processing or equipment failure.
- GS8.**
- Team Coordination & Communication: -
 - Collaborate with collection center workers, processors, and quality assurance teams.
 - Communicate findings and protocols clearly and effectively to ensure smooth operations.
- GS9.** Follow Good Manufacturing Practices (GMP), Good Hygiene Practices (GHP), and Hazard Analysis and Critical Control Point (HACCP) standards. Maintain safety and cleanliness of self, workplace, and equipment.
- GS10.** use the simple IT tools for data recording and reporting can enhance accuracy and efficiency in validation and processing tasks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Hygienic milk production and milk pricing policies</i>	9	10	-	8
PC1. Milk collection and pricing system based on fat content, volume, total milk solids, species of milk animals and cost of milk production	3	2	-	1
PC2. Clean milk production practices in dairy farm	2	3	-	3
PC3. Hygienic production of milk by inspecting raw milk for fulfilment to physical, chemical, and microbiological parameters.	2	3	-	2
PC4. Milk preservation system.	2	2	-	2
<i>Milk procurement and processing operations with complete documentation</i>	21	30	-	22
PC5. Sampling and lactometer reading of procured milk to determine quality following standard operating protocols.	1	3	-	2
PC6. Platform tests (COB, alcohol, sediment, organoleptic evaluation etc.) related to appearance, smell, acidity, consistency, color, temperature etc.	4	6	-	4
PC7. Filtration and checking microbial quality of milk according to standard protocol.	2	2	-	2
PC8. Process of separation through separator to separate cream from milk according to standard protocol.	2	2	-	2
PC9. Homogenization of milk in homogenizer to obtain desirable size of milk fat globule according to standard protocol.	2	2	-	2
PC10. Milk pasteurization according to standard protocol.	2	4	-	2
PC11. Milk cooling in chilling tank until packing.	2	2	-	2
PC12. Standardized and flavored milks (fruit, chocolate etc.) following standard protocol.	2	3	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. CIP cleaning of valves, pipes and pipeline circuits with proper process operations after product manufacture.	2	3	-	2
PC14. Documentation of processed milk related to parameters analyzed, milk storage conditions, process of analysis, equipment usage etc.	2	3	-	2
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1201
NOS Name	Understand and Execute Collection, Validation, Processing and Storage of Raw Milk
Sector	Agriculture
Sub-Sector	
Occupation	Dairy Farm Management
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/AGR/N1202: Apply Dairy Starters and Execute Microbiological Analysis of Milk and Milk Products

Description

This NOS covers the application and management of dairy starter cultures and the execution of microbiological analyses to ensure the safety and quality of milk and its products. It includes understanding the roles and types of dairy starters, ensuring their purity and activity, and propagating them under hygienic conditions. The NOS also involves preparing culture media, isolating and identifying microbial colonies, conducting Grams staining, biochemical tests, and microbial enumeration techniques to detect contamination and maintain microbiological standards in dairy processing.

Scope

The scope covers the following :

- This NOS focuses on the selection, propagation, and preservation of dairy starters used in fermentation processes, along with the execution of microbiological analysis to ensure the safety and quality of milk and dairy products. It covers starter purity checks, activity tests, and sterile propagation techniques. Additionally, it includes microbial culturing, colony isolation, Gram staining, biochemical analysis, and enumeration techniques. The scope ensures the supervisor can detect contamination, identify desirable cultures, and maintain hygienic standards, enabling consistent production of high-quality fermented dairy products.

Elements and Performance Criteria

Selection and importance of dairy starters

To be competent, the user/individual on the job must be able to:

- PC1.** Concept and role of dairy starters in dairy and food fermentations
- PC2.** Starter culture purity check: Gram's staining and catalase test
- PC3.** Starter activity check: Dye reduction tests
- PC4.** Dairy starter cultures propagation under stern hygienic conditions
- PC5.** Preservation of dairy starters

Microbiological analysis of milk and milk products

To be competent, the user/individual on the job must be able to:

- PC6.** Preparation and sterilization of agar media/broth (specific (MRS/M17/PDA/Mac-Conkey)/non-specific (Nutrient agar) with other pre-requisites as per standard operating procedure
- PC7.** Serial dilutions with the sample as per standard protocol
- PC8.** Aseptic techniques (streak/spread/pour plate) for diluted sample in prepared media plates to isolate single colonies
- PC9.** Identify various morphological characteristics of selected, desirable isolated cultures
- PC10.** Gram's staining for microscopical examination of the specific culture
- PC11.** Biochemical analysis of isolated dairy cultures
- PC12.** Preservation selected pure cultures in appropriate broth/glycerol for long term use

Qualification Pack

PC13. Microbial enumeration techniques of dairy foods using specific microbial medium

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the fundamental role of dairy starters in fermentation processes, their types, and their importance in enhancing the taste, texture, and preservation of fermented dairy products.
- KU2.** the key indicators of starter culture purity and activity, including Gram's staining, catalase test, and dye reduction tests to evaluate their efficacy.
- KU3.** hygienic propagation methods of dairy starters and preservation techniques such as freeze drying and spray drying, which ensure long-term viability of cultures.
- KU4.** theoretical understanding of probiotics and related concepts, including their application in developing functional fermented dairy products for health benefits.
- KU5.** microbiology related to dairy, including sample preparation, serial dilution, media preparation, and aseptic techniques like streak/spread/pour plating.
- KU6.** how to prepare and sterilize both specific (MRS, M17, PDA, MacConkey) and non-specific (nutrient agar) media, and understand their use for isolating and cultivating microbial cultures.
- KU7.** identifying morphological characteristics of microbial colonies, and performing biochemical tests to differentiate among dairy-relevant microorganisms.
- KU8.** the function and operation of instruments such as autoclaves, incubators, laminar air flow units, microscopes, and UV-Visible spectrophotometers, essential for microbiological analysis.
- KU9.** microbial enumeration techniques and apply appropriate microbial media to quantify microbial load in milk and milk products.
- KU10.** SOPs, laboratory safety protocols, proper storage of chemicals/media, and the ability to document microbial testing results effectively.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** understand the role, types, and application of dairy starters in fermented dairy product manufacturing, including knowledge of probiotics, prebiotics, synbiotics, and strategies to mitigate starter culture failures.
- GS2.** operate essential lab equipment like autoclaves, laminar airflow units, incubators, microscopes, and use aseptic techniques such as streaking, pouring, and spreading methods for isolating dairy microbes.
- GS3.** Perform starter culture purity checks (e.g., Gram staining, catalase test), activity checks (e.g., dye reduction tests), and microbial enumeration using standard media (MRS, M17, PDA, MacConkey, Nutrient Agar).
- GS4.** Apply preservation methods like freeze-drying or glycerol storage for long-term use of pure dairy cultures, ensuring microbial viability and quality over time.
- GS5.** follow safety guidelines, and maintaining sterile and hygienic lab and production environments, essential for food safety.



Qualification Pack

- GS6.** identify microbial contaminants, analyze culture characteristics, and implement preventive measures to ensure consistent product quality and compliance with food safety norms.
- GS7.** maintain detailed records of microbial tests, results, preservation methods, and process parameters. This supports traceability and regulatory audits.
- GS8.** communicate results and collaborate with dairy technologists, quality analysts, and supervisors effectively, using proper technical language and professional etiquette.
- GS9.** implement food safety regulations like HACCP and FSSAI standards, and understanding of environmental impacts due to poor microbial practices in the dairy industry.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Selection and importance of dairy starters</i>	10	12	-	11
PC1. Concept and role of dairy starters in dairy and food fermentations	2	-	-	2
PC2. Starter culture purity check: Gram's staining and catalase test	2	3	-	2
PC3. Starter activity check: Dye reduction tests	2	3	-	3
PC4. Dairy starter cultures propagation under stern hygienic conditions	2	3	-	2
PC5. Preservation of dairy starters	2	3	-	2
<i>Microbiological analysis of milk and milk products</i>	20	28	-	19
PC6. Preparation and sterilization of agar media/broth (specific (MRS/M17/PDA/Mac-Conkey)/non-specific (Nutrient agar) with other pre-requisites as per standard operating procedure	3	3	-	2
PC7. Serial dilutions with the sample as per standard protocol	1	2	-	1
PC8. Aseptic techniques (streak/spread/pour plate) for diluted sample in prepared media plates to isolate single colonies	2	4	-	2
PC9. Identify various morphological characteristics of selected, desirable isolated cultures	2	3	-	2
PC10. Gram's staining for microscopical examination of the specific culture	2	3	-	2
PC11. Biochemical analysis of isolated dairy cultures	4	5	-	4
PC12. Preservation selected pure cultures in appropriate broth/glycerol for long term use	3	3	-	2
PC13. Microbial enumeration techniques of dairy foods using specific microbial medium	3	5	-	4
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1202
NOS Name	Apply Dairy Starters and Execute Microbiological Analysis of Milk and Milk Products
Sector	Agriculture
Sub-Sector	
Occupation	Dairy Farm Management
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/AGR/N1203: Process Different Types of Dairy Products

Description

The NOS "Process different types of dairy products" (CUTM/AGR/N1202) under the qualification "Dairy Processing and Development Supervisor" (NSQF Level 5) focuses on the production of a wide variety of dairy products. This includes preparing Indian dairy products such as khoa, chhana, paneer, fermented items like dahi and lassi, and fat-rich products like butter and ghee. The NOS emphasizes adherence to hygiene and standard operating procedures during processing, proper utilization of by-products, packaging, storage, and documentation. The aim is to ensure the production of high-quality, safe, and market-ready dairy products through practical application of dairy technologies and quality control measures.

Scope

The scope covers the following :

- The scope of the NOS Process different types of dairy products under the Dairy Processing and Development Supervisor qualification (NSQF Level 5) involves the preparation, production, quality assurance, and documentation of various traditional and value-added dairy products. This includes khoa and its sweets, chhana-based items, fermented products like dahi and lassi, cereal-based products, and fat-rich products like butter and ghee. The NOS also encompasses utilization of by-products, packaging, cold storage, and maintaining product quality through standard operating procedures. Emphasis is placed on hygiene, process standardization, and compliance with safety protocols.

Elements and Performance Criteria

Processing and sustaining quality in manufacturing dairy products

To be competent, the user/individual on the job must be able to:

- PC1.** Indian dairy products and their classification
- PC2.** Preparation and manufacture of khoa
- PC3.** Preparation and manufacture of khoa based sweets (burfi, peda, kalakand, rabidi, milk cake etc.)
- PC4.** Preparation and manufacture of channa
- PC5.** Preparation and manufacture of channa based sweets (gulab jamun, rasomalai, rasogolla, rajbhog etc.)
- PC6.** Preparation of paneer and yield calculations
- PC7.** Preparation of fermented dairy products (dahi, lassi and mishti dahi)
- PC8.** Preparation of cereal based dairy products (kheer and payasam)
- PC9.** Cream and cream standardization
- PC10.** Production of butter
- PC11.** Production of ghee from cream and butter
- PC12.** Utilization of butter milk and ghee residues
- PC13.** Accomplish packaging and proper storage of prepared dairy products

Qualification Pack

- PC14.** Data documentation for type of milk used, milk received from milk collection centers, BMCs, chilling centers etc., information on raw materials, packaging materials, temperature records of cold room/ storage tanks, pasteurizer, chillers and packaging machine
- PC15.** Collect and study market information of dairy products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the types and groups of Indian dairy products such as fat-rich, khoa-based, chhana-based, fermented, and cereal-based products along with their classifications and characteristics.
- KU2.** standard protocols for preparation of various products like khoa, paneer, chhana, and their derivative sweets, fermented items (like dahi, lassi), and fat-rich products like butter and ghee.
- KU3.** the types of milk, coagulants (like citric acid), starter cultures, sugar, flavors, cereals (e.g., basmati rice), and other ingredients used in dairy product manufacturing.
- KU4.** machinery and equipment such as the kova machine, butter churner, ghee settling tank, incubators, sealing and packaging machines, and their cleaning protocols.
- KU5.** cleaning procedures, sanitation standards, disposal of dairy sludge/wastewater, and Good Manufacturing Practices (GMP) essential for ensuring safety in processing environments.
- KU6.** quality control aspects such as texture, taste, aroma, microbial testing, and record-keeping of processed dairy products.
- KU7.** appropriate packaging materials, temperature requirements, and storage conditions to maintain the shelf life and safety of dairy products.
- KU8.** methods to utilize byproducts such as buttermilk and ghee residues, minimizing waste and maximizing output efficiency.
- KU9.** data related to milk source, raw materials used, equipment parameters, and environmental conditions during processing and storage.
- KU10.** consumer preferences, market trends, and value-added product development to enhance product diversification and profitability.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow production protocols for a wide range of dairy products including khoa, chhana, paneer, fermented products (lassi, dahi), cereal-based products (kheer, payasam), butter, and ghee.
- GS2.** Hands-on experience with equipment such as kova machines, butter churners, cream separators, packaging and sealing machines, ensuring efficiency and hygiene during processing.
- GS3.** follow standardized production protocols for each type of product including ingredient proportions, processing time, and temperature controls.
- GS4.** conduct product quality checks based on appearance, aroma, taste, and document quality control and microbial test records to maintain industry standards.
- GS5.** maintain documentation data like milk sources, raw materials, temperature logs, equipment usage, and market information for traceability and process improvement.



Qualification Pack

- GS6.** to plan production schedules, coordinate with suppliers for ingredients, and manage storage and packaging operations while maintaining freshness and shelf-life.
- GS7.** do safe disposal of dairy sludge, residues, wastewater, and ensuring compliance with environmental standards during product processing.
- GS8.** communicate effectively with team members, supervisors, and stakeholders, and working collaboratively in a clean, safe, and professional environment.
- GS9.** collect and interpret market trends, consumer preferences, and adapt production accordingly, fostering continuous improvement and innovation.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Processing and sustaining quality in manufacturing dairy products</i>	30	40	-	30
PC1. Indian dairy products and their classification	2	2	-	2
PC2. Preparation and manufacture of khoa	2	2	-	2
PC3. Preparation and manufacture of khoa based sweets (burfi, peda, kalakand, rabidi, milk cake etc.)	2	3	-	2
PC4. Preparation and manufacture of channa	2	3	-	2
PC5. Preparation and manufacture of channa based sweets (gulab jamun, rasomalai, rasogolla, rajbhog etc.)	2	3	-	2
PC6. Preparation of paneer and yield calculations	2	2	-	2
PC7. Preparation of fermented dairy products (dahi, lassi and mishti dahi)	3	4	-	3
PC8. Preparation of cereal based dairy products (kheer and payasam)	2	2	-	2
PC9. Cream and cream standardization	2	3	-	2
PC10. Production of butter	2	4	-	2
PC11. Production of ghee from cream and butter	2	3	-	2
PC12. Utilization of butter milk and ghee residues	2	3	-	2
PC13. Accomplish packaging and proper storage of prepared dairy products	2	3	-	2
PC14. Data documentation for type of milk used, milk received from milk collection centers, BMCs, chilling centers etc., information on raw materials, packaging materials, temperature records of cold room/ storage tanks, pasteurizer, chillers and packaging machine	2	2	-	2



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Collect and study market information of dairy products	1	1	-	1
NOS Total	30	40	-	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1203
NOS Name	Process Different Types of Dairy Products
Sector	Agriculture
Sub-Sector	
Occupation	Dairy Farm Management
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/AGR/N1204: Perform Chemical Quality Testing of Raw Milk, Processed Milk and Milk Products

Description

This NOS covers the skills and knowledge required to carry out chemical quality testing of raw milk, processed milk, and various dairy products in a dairy processing unit. It includes preparing and handling testing equipment, conducting chemical tests to determine parameters like fat, SNF, acidity, and presence of adulterants, and accurately recording and reporting results. The role also involves adhering to food safety norms, laboratory hygiene practices, and standard operating procedures (SOPs) to ensure product quality and regulatory compliance.

Scope

The scope covers the following :

- This NOS covers the competencies required to carry out chemical quality testing of raw milk, processed milk, and various milk products. It includes sample collection, preparation, and testing using standard methods and equipment, while maintaining hygiene, accuracy, and safety protocols. The role ensures compliance with food safety standards by identifying chemical parameters such as fat content, SNF, acidity, and presence of adulterants. The scope also includes documentation of test results and coordination with relevant quality assurance teams for corrective actions.

Elements and Performance Criteria

Chemical quality checking of raw and processed milk

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate preparation and standardization of dairy chemical reagents.
- PC2.** Calculate density and specific gravity of milk at a fixed temperature by using standard formula.
- PC3.** Calculate viscosity and surface tension of milk at a fixed temperature by using standard formula.
- PC4.** Impact of essential chemical parameters such as pH, titratable acidity and peroxide value on milk quality.
- PC5.** Detect the presence of sugar in milk by resorcinol test.
- PC6.** Detect the presence of starch in milk by iodine test.
- PC7.** Detect the presence of salt in milk.
- PC8.** Detect the presence of neutralizer in milk by rosolic acid test.
- PC9.** Detect the presence of detergent in milk by methylene blue reduction test.
- PC10.** Fat content in pasteurized milk by using Gerber method or using a milk analyzer.
- PC11.** Solids-not-fat content in pasteurized milk by using lactometer or using a milk analyzer.
- PC12.** Phosphatase test to check the pasteurization efficiency.

Chemical quality checking of processed dairy products

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC13.** Moisture and fat content in ghee and butter using standard methodology.
- PC14.** Salt and curd content in butter using standard methodology.
- PC15.** Free fatty acid content in ghee using standard methodology.
- PC16.** Fat, moisture and protein content of chhana and paneer using standard methodology.
- PC17.** Fat, moisture and protein content of khoa using standard methodology.
- PC18.** Acidity and pH of dahi and lassi using standard methodology.
- PC19.** Fat, protein and lactose content of dahi and lassi using standard methodology.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the chemical composition of raw milk and processed dairy products including fat, protein, SNF, lactose, and minerals.
- KU2.** chemical testing methods such as titration, Gerber method, alcohol test, acidity test, neutralizer test, and total solids measurement.
- KU3.** various lab instruments like butyrometers, lactometers, burettes, pipettes, and their calibration and maintenance protocols.
- KU4.** hygienic practices, sample handling procedures, contamination prevention, and safe disposal of chemical reagents and samples.
- KU5.** FSSAI norms, BIS standards, and other legal parameters for milk quality testing and chemical analysis.
- KU6.** recording test results, maintaining logbooks, ensuring traceability, and reporting deviations in standard formats.
- KU7.** and interpret chemical test outcomes to determine compliance with quality benchmarks and specifications.
- KU8.** how microbial contamination may influence chemical test results and product quality.
- KU9.** chemical hazards, handling of acids and alkalis, use of PPE, and first aid response in case of spills or exposure.
- KU10.** importance in maintaining consistency, accuracy, and safety during testing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** clearly communicate test procedures, findings, and quality issues with team members and supervisors, both verbally and in writing.
- GS2.** perform calculations and interpreting results related to chemical tests such as fat, SNF (Solids Not Fat), acidity, and adulteration.
- GS3.** operate digital instruments and laboratory equipment like milk analyzers, pH meters, and using software to log and analyze test results.
- GS4.** identify deviations from quality standards and make informed decisions regarding product acceptability or further testing.



Qualification Pack

- GS5.** maintain accurate records of test results, observations, and sample tracking for traceability and audit purposes.
- GS6.** troubleshoot inconsistencies in test results and identify the causes of chemical quality issues.
- GS7.** Work effectively as part of a team in a laboratory or production environment to ensure quality objectives are met.
- GS8.** handle chemicals and biological samples, and maintaining hygienic conditions during quality testing.
- GS9.** to learn new testing methods, instruments, and quality standards in line with industry advancements and FSSAI norms.
- GS10.** Understand of consumer safety, product compliance, and regulatory standards (e.g., BIS, FSSAI) relevant to milk and milk product testing.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Chemical quality checking of raw and processed milk</i>	18	26	-	18
PC1. Demonstrate preparation and standardization of dairy chemical reagents.	2	2	-	2
PC2. Calculate density and specific gravity of milk at a fixed temperature by using standard formula.	2	2	-	2
PC3. Calculate viscosity and surface tension of milk at a fixed temperature by using standard formula.	2	2	-	2
PC4. Impact of essential chemical parameters such as pH, titratable acidity and peroxide value on milk quality.	2	3	-	2
PC5. Detect the presence of sugar in milk by resorcinol test.	1	2	-	1
PC6. Detect the presence of starch in milk by iodine test.	1	2	-	1
PC7. Detect the presence of salt in milk.	1	2	-	1
PC8. Detect the presence of neutralizer in milk by rosolic acid test.	1	2	-	1
PC9. Detect the presence of detergent in milk by methylene blue reduction test.	2	2	-	1
PC10. Fat content in pasteurized milk by using Gerber method or using a milk analyzer.	2	3	-	2
PC11. Solids-not-fat content in pasteurized milk by using lactometer or using a milk analyzer.	1	2	-	1
PC12. Phosphatase test to check the pasteurization efficiency.	1	2	-	2
<i>Chemical quality checking of processed dairy products</i>	12	14	-	12
PC13. Moisture and fat content in ghee and butter using standard methodology.	2	2	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Salt and curd content in butter using standard methodology.	2	2	-	2
PC15. Free fatty acid content in ghee using standard methodology.	1	2	-	2
PC16. Fat, moisture and protein content of chhana and paneer using standard methodology.	2	2	-	2
PC17. Fat, moisture and protein content of khoa using standard methodology.	1	2	-	1
PC18. Acidity and pH of dahi and lassi using standard methodology.	2	2	-	1
PC19. Fat, protein and lactose content of dahi and lassi using standard methodology.	2	2	-	2
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1204
NOS Name	Perform Chemical Quality Testing of Raw Milk, Processed Milk and Milk Products
Sector	Agriculture
Sub-Sector	
Occupation	Dairy Farm Management
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/VSQ/N7804: Ensure organizational health, safety and security for self and others

Description

This NOS focuses on maintaining personal hygiene, workplace safety, and environmental responsibility. It outlines procedures for disease prevention, safe equipment handling, waste disposal, and emergency response. The standard ensures that individuals follow preventive measures and safety protocols to create a clean, hazard-free, and health-conscious work environment.

Scope

The scope covers the following :

- The scope covers adherence to sanitation and hygiene norms, safe use of tools and chemicals, proper waste management, first aid administration, and compliance with government and workplace advisories. It aims to prepare individuals to work safely under routine and emergency situations, reducing risks to themselves, coworkers, and the environment.

Elements and Performance Criteria

Maintain personal hygiene

To be competent, the user/individual on the job must be able to:

- PC1.** wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals
- PC2.** wash the worn clothes with soap and sundry before use next time
- PC3.** ensure the face is covered with mask or three layers of cloth-piece
- PC4.** follow the workplace sanitization norms including distancing from sick people

Maintain clean and safe workplace

To be competent, the user/individual on the job must be able to:

- PC5.** carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor
- PC6.** follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards
- PC7.** sanitize equipment, tools and machinery before and after use
- PC8.** dispose waste safely and correctly in the designated area
- PC9.** work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed
- PC10.** follow government / workplace advisories in case of outbreak of any disease/disaster

Administer appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements

Qualification Pack

- PC12.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques
- PC13.** report details of first aid administered in accordance with workplace procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of personal hygiene in preventing infection and illness.
- KU2.** Use and effectiveness of masks and sanitizers.
- KU3.** Workplace safety norms, including sanitization and distancing practices.
- KU4.** Basic safety checks and hazard identification protocols.
- KU5.** Safe handling and usage instructions of chemicals and pesticides.
- KU6.** Procedures for cleaning and sanitizing tools and machinery.
- KU7.** Waste disposal methods and designated disposal areas.
- KU8.** Environmental safety practices and risk mitigation.
- KU9.** Government and workplace advisories related to health emergencies.
- KU10.** First aid techniques and workplace protocols for injury reporting.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to follow hygiene and sanitization routines consistently.
- GS2.** Situational awareness to identify unsafe practices or potential hazards.
- GS3.** Communication skills to report hazards and emergencies accurately.
- GS4.** Decision-making during accidents or outbreaks.
- GS5.** Capability to use PPE and follow safety signs/instructions.
- GS6.** Time management in emergency response or first aid delivery.
- GS7.** Teamwork while maintaining workplace health and safety norms.
- GS8.** Basic technical skill to perform equipment safety checks.
- GS9.** Problem-solving in waste management and chemical handling.
- GS10.** Record-keeping for first aid and incident reporting.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal hygiene</i>	10	20	-	-
PC1. wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals	2	5	-	-
PC2. wash the worn clothes with soap and sundry before use next time	2	5	-	-
PC3. ensure the face is covered with mask or three layers of cloth-piece	2	5	-	-
PC4. follow the workplace sanitization norms including distancing from sick people	4	5	-	-
<i>Maintain clean and safe workplace</i>	10	34	-	-
PC5. carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor	2	6	-	-
PC6. follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards	2	5	-	-
PC7. sanitize equipment, tools and machinery before and after use	2	6	-	-
PC8. dispose waste safely and correctly in the designated area	2	6	-	-
PC9. work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed	1	6	-	-
PC10. follow government / workplace advisories in case of outbreak of any disease/disaster	1	5	-	-
<i>Administer appropriate emergency procedures</i>	10	16	-	-
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements	2	6	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques	2	5	-	-
PC13. report details of first aid administered in accordance with workplace procedures	6	5	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7804
NOS Name	Ensure organizational health, safety and security for self and others
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



Qualification Pack

3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 60% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

Qualification Pack

whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N1201.Understand and Execute Collection, Validation, Processing and Storage of Raw Milk	30	40	0	30	100	20
CUTM/AGR/N1202.Apply Dairy Starters and Execute Microbiological Analysis of Milk and Milk Products	30	40	0	30	100	20
CUTM/AGR/N1203.Process Different Types of Dairy Products	30	40	0	30	100	20
CUTM/AGR/N1204.Perform Chemical Quality Testing of Raw Milk, Processed Milk and Milk Products	30	40	0	30	100	20
CUTM/VSQ/N7804.Ensure organizational health, safety and security for self and others	30	70	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	170	260	-	120	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
QF	Qualification File
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
TVET	Technical and Vocational Education and Training
SOP	Standard Operating Procedure
OH&S	Occupational Health and Safety
PPE	Personal Protective Equipment
HACCP	Hazard Analysis and Critical Control Points
GMP	Good Manufacturing Practices
FSSAI	Food Safety and Standards Authority of India
COB	Clot on Boiling
SNF	Solid Not Fat
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
QF	Qualification File
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
TVET	Technical and Vocational Education and Training
SOP	Standard Operating Procedure
OH&S	Occupational Health and Safety
PPE	Personal Protective Equipment
HACCP	Hazard Analysis and Critical Control Points
GMP	Good Manufacturing Practices
FSSAI	Food Safety and Standards Authority of India
COB	Clot on Boiling
SNF	Solid Not Fat

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above.
Declarative Knowledge	Declarative knowledge refers to facts, concepts and principles that need to be known and/or understood in order to accomplish a task or to solve a problem.
Key Learning Outcome	Key learning outcome is the statement of what a learner needs to know, understand and be able to do in order to achieve the terminal outcomes. A set of key learning outcomes will make up the training outcomes. Training outcome is specified in terms of knowledge, understanding (theory) and skills (practical application).

Qualification Pack

OJT (M)	On-the-job training (Mandatory) trainees are mandated to complete specified hours of training on site
OJT (R)	On-the-job training (Recommended) trainees are recommended the specified hours of training on site
Procedural Knowledge	Procedural knowledge addresses how to do something, or how to perform a task. It is the ability to work, or produce a tangible work output by applying cognitive, affective or psychomotor skills.
Training Outcome	Training outcome is a statement of what a learner will know, understand and be able to do upon the completion of the training.
Terminal Outcome	Terminal outcome is a statement of what a learner will know, understand and be able to do upon the completion of a module. A set of terminal outcomes help to achieve the training outcome.
Declarative Knowledge	Declarative knowledge refers to facts, concepts and principles that need to be known and/or understood in order to accomplish a task or to solve a problem.
Key Learning Outcome	Key learning outcome is the statement of what a learner needs to know, understand and be able to do in order to achieve the terminal outcomes. A set of key learning outcomes will make up the training outcomes. Training outcome is specified in terms of knowledge, understanding (theory) and skills (practical application).
OJT (M)	On-the-job training (Mandatory) trainees are mandated to complete specified hours of training on site
OJT (R)	On-the-job training (Recommended) trainees are recommended the specified hours of training on site
Procedural Knowledge	Procedural knowledge addresses how to do something, or how to perform a task. It is the ability to work, or produce a tangible work output by applying cognitive, affective or psychomotor skills.
Training Outcome	Training outcome is a statement of what a learner will know, understand and be able to do upon the completion of the training.
Terminal Outcome	Terminal outcome is a statement of what a learner will know, understand and be able to do upon the completion of a module. A set of terminal outcomes help to achieve the training outcome.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.

Qualification Pack

Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf